Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A transformer has an efficiency of 75%. The power input is $4\ \mathrm{kW}$ at $100\ \mathrm{V}$. If the secondary voltage is $200\ \mathrm{V}$, calculate the currents in the primary and secondary.

(A) 40 A, 15 A

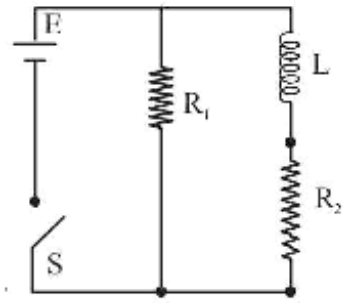
(B) 20 A,18 A

(C) 30 A,16 A

(D) 30A, 24 A

Q.2

An inductor of inductance $L = 400\mathrm{mH}$ and resistors of resistance $R_1 = 2\Omega$ and $\mathrm{R}_2 = 2\Omega$ are connected to a battery of emf $12\ \mathrm{V}$ as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at $t = 0$. The potential drop across L as a function of time is



(A) $\frac{12}{t}e^{-3t}\ \mathrm{V}$

(B) $6\left(1 - e^{-t/0.2}\right)\ \mathrm{V}$

(C) $12\mathrm{e}^{-5t}\ \mathrm{V}$

(D) $6\mathrm{e}^{-5t}\ \mathrm{V}$

Q.3

Combination of two identical capacitors, a resistor R and $a\ dc$ voltage source of voltage $6\ \mathrm{V}$ is used in an experiment on a $(C - R)$ circuit. It is found that for a parallel combination of the capacitor the time in which the voltage of the fully charged combination reduces to half its original voltage is 10 second. For series combination, what is the time (in second) needed for reducing the voltage of the fully charged series combination by half?

(A) 2.0 sec

(B) 2.5 sec

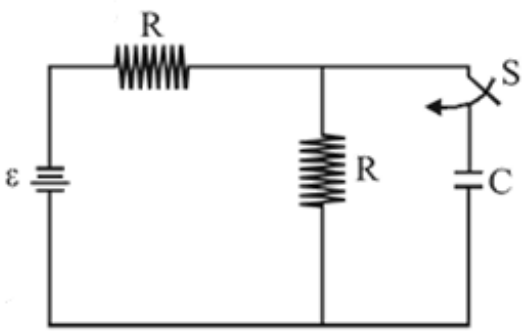
(C) 2.3 sec

(D) 2.7 sec

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Q.4

In given circuit capacitor is initially uncharged. Now at $\mathrm{t} = 0$ switch S is closed then current given by source at any time t is



(A) $\frac{2}{\mathrm{R}}\left(1-\mathrm{e}^{\frac{-2t}{\mathrm{CR}}}\right)$

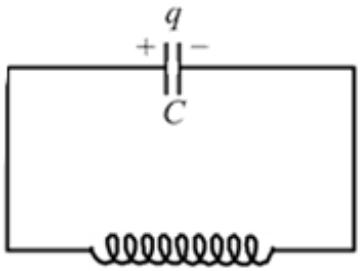
(B) $\frac{\varepsilon}{2R}\left(1+\mathrm{e}^{\frac{-2t}{\mathrm{CR}}}\right)$

(C) $\frac{\varepsilon}{2R}\left(1-\mathrm{e}^{\frac{-2t}{\mathrm{CR}}}\right)$

(D) $\frac{2\varepsilon}{\mathrm{R}}\left(1-\mathrm{e}^{\frac{-2t}{\mathrm{CR}}}\right)$

Q.5

In an $L-C$ circuit shown in the figure, $C = 1\ \mathrm{F}, L = 4\mathrm{H}$. At time $t = 0$, charge in the capacitor is $4C$ and it is decreasing at a rate of $\sqrt{5}\mathrm{C/s}$. Choose the correct statements.



(A) maximum charge in the capacitor can be $6\mathrm{C}$

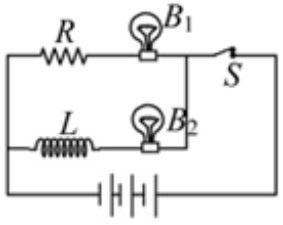
(B) maximum charge in the capacitor can be $8\mathrm{C}$

(C) charge in the capacitor will be maximum after time $2\sin^{-1}(2/3)\mathrm{sec}$

(D) None of these

Q.6

Figure shows two bulbs B_1 and B_2, resistor R and an inductor L. When the switch S is turned off



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(A) both B_1 and B_2 die out promptly

(B) both B_1 and B_2 die out with some delay

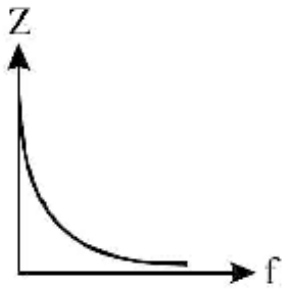
(C) B_1 dies out promptly but B_2 with some delay

(D) B_2 dies out promptly but B_1 with some delay

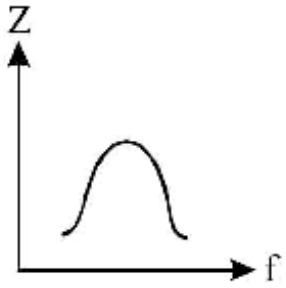
Q.7

Which one of the following curves represents the variation of impedance (Z) with frequency f in series LCR circuit?

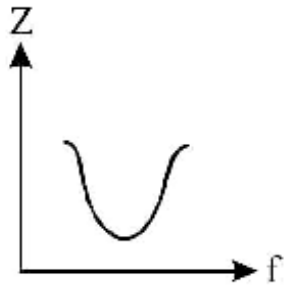
(A)



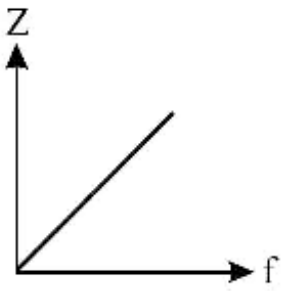
(B)



(C)



(D)



Q.8

The power factor in a circuit connected to an A.C. source. The value of power factor is

(A) unity when the circuit contains an ideal inductance only

(B) unity when the circuit contains an idea resistance only

(C) zero when the circuit contains an idea resistance only

(D) unity when the circuit contains an ideal capacitance only

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Q.9

A coil of inductive reactance 31Ω has a resistance of 8Ω. It is placed in series with a condenser of capacitative reactance 25Ω. The combination is connected to an a.c. source of 110 volt. The power factor of the circuit is

(A) 0.64

(B) 0.80

(C) 0.33

(D) 0.56

Q.10

A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to 2R. A capacitance of capacitive reactance equal to R is added in series with L and R. The ratio of the new power factor to the old one is

(A) $\sqrt{\frac{2}{3}}$

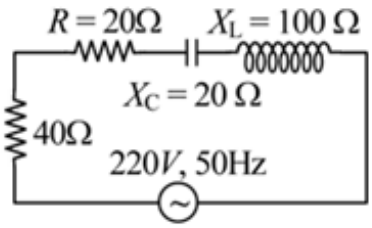
(B) $\sqrt{\frac{2}{5}}$

(C) $\sqrt{\frac{3}{2}}$

(D) $\sqrt{\frac{5}{2}}$

Q.11

The power factor of the circuit shown in the figure is



(A) 0.4

(B) 0.2

(C) 0.8

(D) 0.6

Q.12

The primary winding of a transformer has 100 turns and its secondary winding has 200 turns. The primary is connected to an A.C. supply of $120\ \mathrm{V}$ and the current flowing in it is 10 A. The voltage and the current in the secondary are

(A) $240\ \mathrm{V}, 5\ \mathrm{A}$

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(B) $240\ \text{V}, 10\ \text{A}$

(C) $60\ \text{V}, 20\ \text{A}$

(D) $120\ \text{V}, 20\ \text{A}$

Q.13

The primary and secondary coil of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in webers, t is time in seconds and ϕ_0 is a constant output voltage (in volts) across the secondary coil is?

(A) 140 V

(B) 130 V

(C) 120 V

(D) 110 V

Q.14

The transformer voltage induced in the secondary coil of a transformer is mainly due to

(A) a varying electric field

(B) a varying magnetic field

(C) the vibrations of the primary coil

(D) the iron core of the transformer

Q.15

A step-down transformer transforms a supply line voltage of 2200 volt into 220 volt. The primary coil has 5000 turns. The number of turns in the secondary are

(A) 5000

(B) 550

(C) 500

(D) 450

Answers & Solutions

Q.1 Answer:

40 A, 15 A

Solution:

$P_i = e_p I_p = 4\ \text{kW} = 4000\ \text{W}, e_p = 100\ \text{V}$ **and** $e_s = 200\ \text{V}$

$I_p = \frac{P_i}{e_p} = \frac{4000}{100} = 40\ \text{A}$

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$\eta = \frac{P_o}{P_i} \Rightarrow P_o = \eta P_i = 0.75 \times 4000 = 3000 \text{ W}$
$\therefore \quad I_s = \frac{P_o}{e_s} = \frac{3000}{200} = 15 \text{ A}$

Q.2 Answer:

$12\text{e}^{-5t} \text{ V}$

Solution:

Growth in current in LR_2 branch when switch is closed is given by
$i = \frac{E}{R_2}\left[1 - e^{-R_2 t/L}\right]$
$\Rightarrow \frac{di}{dt} = \frac{E}{R_2} \cdot \frac{R_2}{L} \cdot e^{-R_2 t/L} = \frac{E}{L} e^{-\frac{R_2 t}{L}}$
Hence, potential drop across L
$= \left(\frac{E}{L} e^{-R_2 t/L}\right) L = E e^{-R_2 t/L}$
$= 12 e^{-\frac{2t}{400 \times 10^{-3}}} = 12\text{e}^{-5t} \text{ V}$

Q.3 Answer:

2.5 sec

Solution:

Time constant for parallel combination $= 2RC$
Time constant for series combination
$= \frac{RC}{2}$
In first case :
$V = V_0 e^{-\frac{t_1}{2RC}} = \frac{V_0}{2}$
In second case :
$V = V_0 e^{-\frac{t_2}{(RC/2)}} = \frac{V_0}{2}$
From (1) and (2)
$\frac{t_1}{2RC} = \frac{t_2}{(RC/2)}$
$\Rightarrow t_2 = \frac{t_1}{4} = \frac{10}{4} = 2.5\text{sec}$

Q.4 Answer:

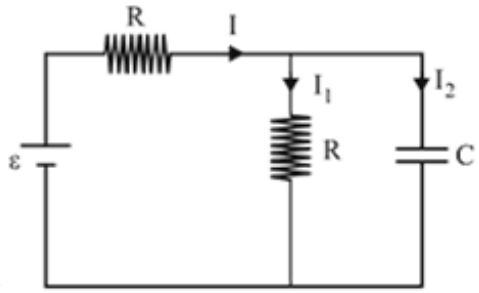
$\frac{\varepsilon}{2R}\left(1 + \text{e}^{\frac{-2t}{\text{CR}}}\right)$

Solution:

$\text{Q} = \frac{\text{EC}}{2}\left(1 - \text{e}^{-\text{t}/\tau}\right)$
$\text{I}_2 = \frac{\varepsilon \text{C}}{2\text{r}} \text{e}^{-\text{t}/\tau} = \frac{\varepsilon}{\text{R}} \text{e}^{-\text{t}/\tau}$
$\text{I}_1 = \frac{\text{Vc}}{\text{R}} = \frac{\varepsilon/2}{\text{R}}\left(1 - \text{e}^{-\text{t}/\tau}\right)$
$\text{I}_1 = \frac{\varepsilon}{\text{R}} - \text{e}^{-\text{t}/\tau} + \frac{\varepsilon}{2\text{R}} - \frac{\varepsilon}{2\text{R}} \varepsilon^{-\text{t}/\tau}$

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$\frac{\varepsilon}{2R}\left(1+\mathrm{e}^{-t/\tau}\right)=\frac{\varepsilon}{2R}\left(1+e^{\frac{-2t}{CR}}\right)$



Q.5 Answer:

maximum charge in the capacitor can be $6\mathrm{C}$

Solution:

$i=\sqrt{5}A$

$\frac{q_m^2}{2C}=\frac{q^2}{2C}+\frac{1}{2}Li^2\Rightarrow q_{\max}=6C$

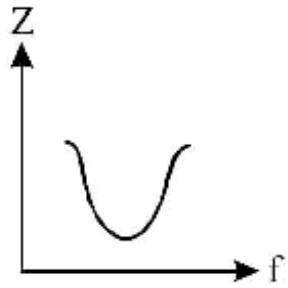
Q.6 Answer:

both B_1 and B_2 die out with some delay

Solution:

In L-R circuit, current decreases exponentially.

Q.7 Answer:



Solution:

Impedance at resonant frequency is minimum in series LCR circuit.

So, $Z=\sqrt{R^2+\left(2\pi\mathrm{fL}-\frac{1}{2\pi\mathrm{fC}}\right)^2}$

Q.8 Answer:

unity when the circuit contains an idea resistance only

Solution:

$\cos\phi=\frac{\mathrm{R}}{\mathrm{Z}}$**, where Z is the impedance &**

$Z=\sqrt{R^2+\left(X_L-X_C\right)^2}$**, if there is only resistance then** $\mathrm{Z}=\mathrm{R}\Rightarrow\cos\phi=1$

Q.9 Answer:

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0.80

Solution:

$$\text{Power factor, } \phi = \frac{R}{\sqrt{\left(\omega L - \frac{1}{\omega C}\right)^2 + R^2}}$$

$$= \frac{8}{\sqrt{(31-25)^2 + 8^2}} \qquad = \frac{8}{\sqrt{6^2 + 8^2}} = \frac{8}{10} = 0.8$$

Q.10 Answer:

$\sqrt{\frac{5}{2}}$

Solution:

$$\text{Power factor (old)} = \frac{R}{\sqrt{R^2 + X_L^2}} = \frac{R}{\sqrt{R^2 + (2R)^2}} = \frac{R}{\sqrt{5}R}$$

$$\text{Power factor (new)} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{R}{\sqrt{R^2 + (2R - R)^2}} = \frac{R}{\sqrt{2}R}$$

$$\frac{\text{New power factor}}{\text{Old power factor}} = \frac{\frac{R}{\sqrt{2}R}}{\frac{R}{\sqrt{5}R}} = \sqrt{\frac{5}{2}}$$

Q.11 Answer:

0.6

Solution:

$$\textbf{Power factor} = \frac{R}{z} = \frac{20+40}{\sqrt{60^2 + (100-20)^2}} = \frac{60}{100} = 0.6 \; (R_{\text{eq}} = 20 + 40 = 60\Omega)$$

Q.12 Answer:

240 V, 5 A

Solution:

$$\frac{\mathrm{E_s}}{\mathrm{E_p}} = \frac{\mathrm{n_s}}{\mathrm{n_p}} \textbf{ or } \mathrm{E_s} = \mathrm{E_p} \times \left(\frac{\mathrm{n_s}}{\mathrm{n_p}}\right)$$

$$\therefore \mathrm{E_s} = 120 \times \left(\frac{200}{100}\right) = 240 \text{ V}$$

$$\frac{\mathrm{I_p}}{\mathrm{I_s}} = \frac{\mathrm{n_s}}{\mathrm{n_p}} \textbf{ or } \mathrm{I_s} = \mathrm{I_p}\left(\frac{\mathrm{n_p}}{\mathrm{n_s}}\right) \qquad \therefore \mathrm{I_s} = 10\left(\frac{100}{200}\right) = 5\text{amp}$$

Q.13 Answer:

120 V

Solution:

Since $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

Where N_s **= No. of turns across primary coil** $= 50$ N_p **= No. of turns across secondary coil** $= 1500$

and $V_p = \frac{d\phi}{dt} = \frac{d}{dt}(\phi_0 + 4t) = 4$

$\Rightarrow V_s = \frac{1500}{50} \times 4 = 120$ V

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Q.14 Answer:

a varying magnetic field

Solution:

Thus, it is the change in magnetic field that induces emf in the secondary coil. So, we can say that the transformer voltage induced in the secondary coil of α transformer is mainly due to a varying magnetic field.

Q.15 Answer:

500

Solution:

$\frac{e_s}{e_p} = \frac{N_s}{N_p}; \frac{220}{2200} = \frac{N_s}{5000}$

$\therefore N_s = 500$

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